A1 → A2 → B1 → B2 → A3 → B3

X3DH → Shared root key
RK0 (= SK

Alice

Alice (init)
DHA0
RK0 - RK → RK1
via DH (DHA0, SPK_B)
send-chain CK-A0
DHr = SPK-B

BOB

BOB (Bundle)
hold SPK-B (DHB0)
RK = RK0
DHr = none

DH RATCHET
A1 heads {dh = DHA0, Pn = 0, n = 0} → MK-A1
RK0 → RK1 → RK2
recv-chain CK-A0 (= Alice send)

DHs = DHB1

SYMMETRIC
A2 heads {dh = DHA0, Pn = 0, n = 1} → MK-A2
same DH key, CKA1 → MK-A2 → CK-A2

DH RATCHET
B1 heads {dh = DHB1, Pn = 0, n = 0} → MK-B1
RK1 → RK2 → RK3
Recv-chain CK-B0 (= Bob's send)

DHs = DHA1

SYMMETRIC
B2 {dh = DHB1, Pn = 0, n = 1} → MK-B2
same DH
CK-B1 → MK-B2 → CK-B2

DH RATCHET
A3 {DHA1, Pn = 2, n = 0} → MK-A3
RK2 → RK3 → RK4
recv-chain CK-A'0 (= Alice send)

DHs = DHB2

DH RATCHET
B3 {DHB2, Pn = 2, n = 0} → MK-B3
RK3 → RK4 → RK5
recv-chain CK-B'0 = (= BOB send)

DHs = DHA2

Alice find
DHs = DHA2
DHr = DHB2
RK = RK5

Bob find
DHs = DHB2
DHr = DHA1
RK = RK4